# CardioLens 3D

Coronary risk studio: CAD and LAD / LCX / RCA stenosis probability on an interactive 3D heart, explained, and agent-ready.

Digital Twin Hackathon · Cardiovascular Risk Visualization & Prediction

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PROBLEM STATEMENT

# Risk is reported as one bare number

- Coronary artery disease is a leading cause of death; invasive angiography is the reference test but is costly and not risk-free.
- Clinical risk scores give one percentage and little insight into which artery is affected or why.
- Clinicians need a transparent estimate per vessel (LAD, LCX, RCA), with reasons and honest limits, before deciding on imaging.

**Our answer**

Per-vessel probability on a 3D heart, exact SHAP explanations, what-if curves, and a scan reading room, all in one file.

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HEALTHCARE USE CASE

# Where it helps in care

## Triage
Rank patients for CT / angiography using routine clinical data.

## Shared decisions
Show a patient which vessel is at risk and what drives it.

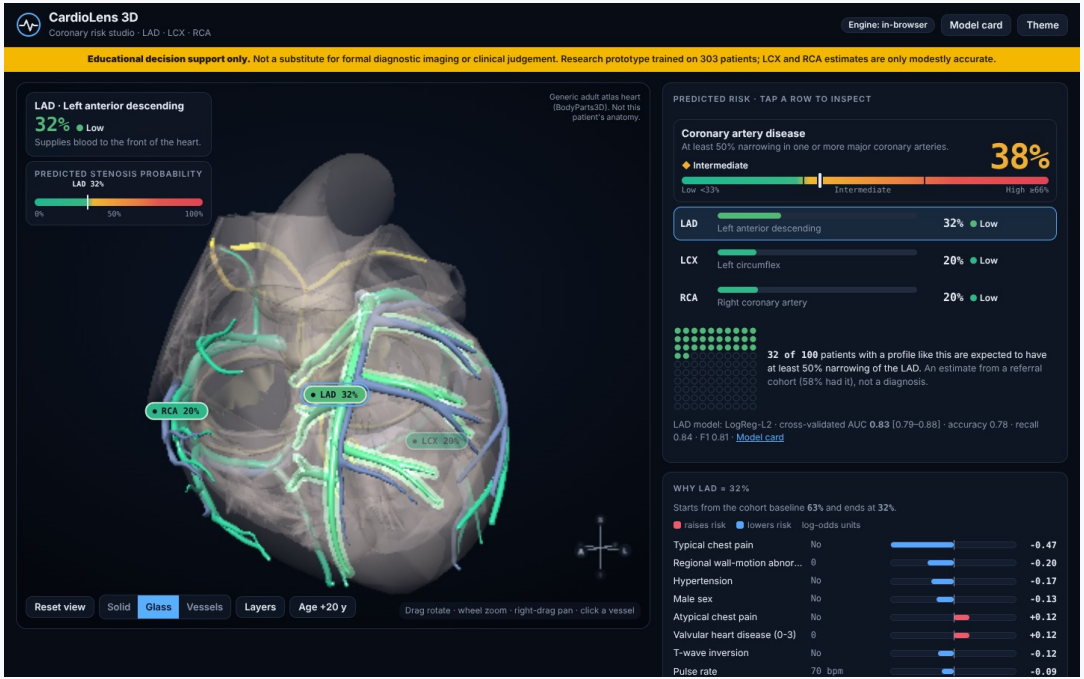
## Teaching
Explore what-if changes (age, diabetes, EF) on a 3D heart.

## Imaging support
Calcium-candidate finder on CT; LV volumes and EF from cine MRI.

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THE SOLUTION

# One page: prediction, 3D heart, explanation



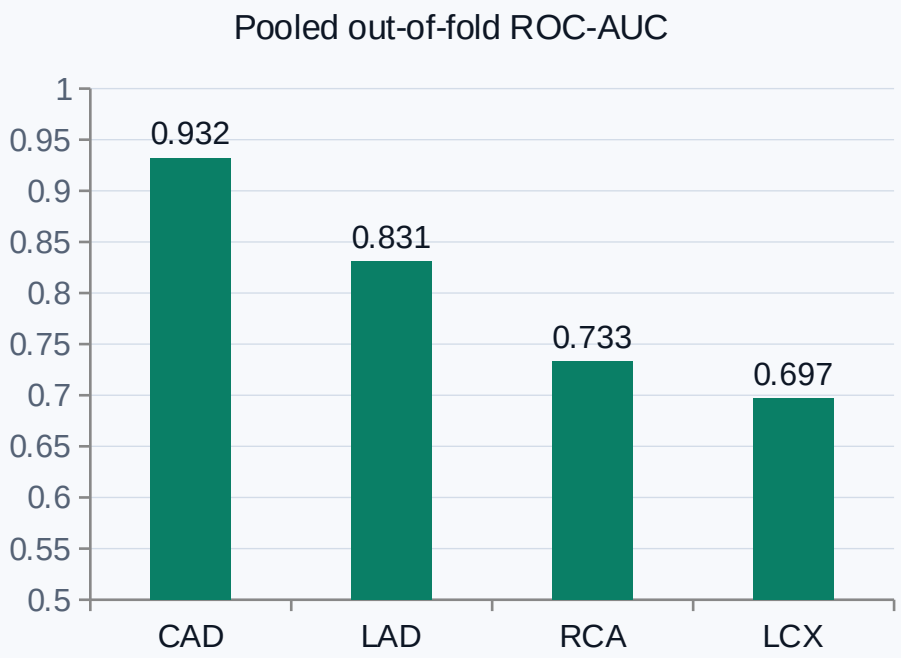

- Vessel colour = predicted probability (teal → amber → red) plus text and icon
- Click a vessel to fly to it; Solid / Glass / Vessels; Age +20 y
- Runs in the browser, no GPU, no upload

Self-contained HTML · works offline

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AI / ML MODEL

# Nested cross-validation, honest results

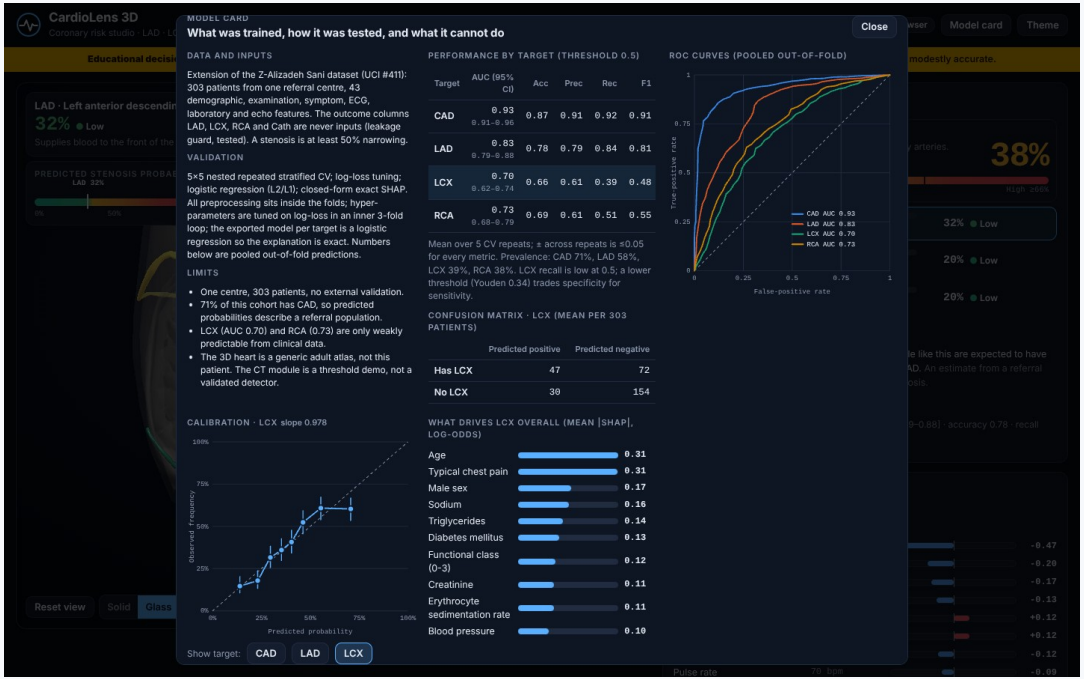



- Extension of Z-Alizadeh Sani (UCI #411): 303 patients, 43 inputs
- 7 model families, 5×5 repeated stratified nested CV, log-loss tuning
- Exported: logistic regression (L2; L1 for RCA) → exact SHAP, runs in the browser
- Outcome columns never used as inputs (leakage guard); P(vessel) ≤ P(CAD)
- JS engine equals scikit-learn to 5.6e-16

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CLINICAL INTERPRETABILITY

# Why this probability?

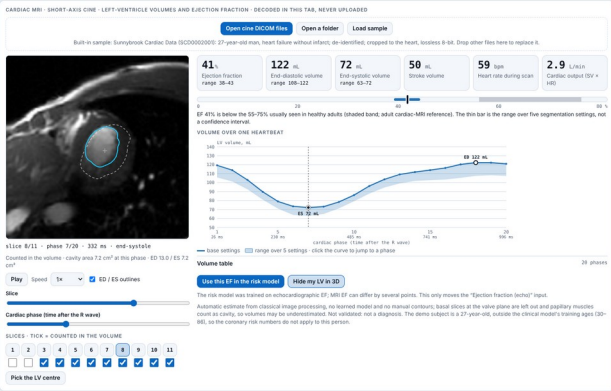


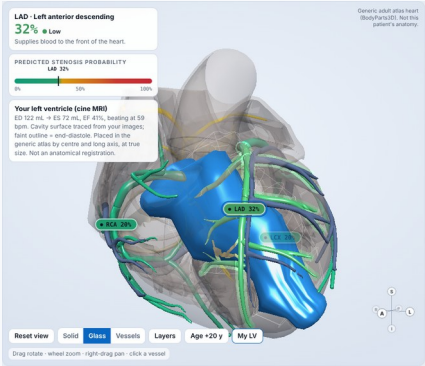

- Exact SHAP bars per patient and per clinical domain
- What-if curves; out-of-range region hatched
- "n of 100 patients like this"
- Model card: ROC, calibration, confusion matrix, limits
- Disclaimer banner always visible

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MULTIMODAL

# Scan reading room: CT and cardiac MRI



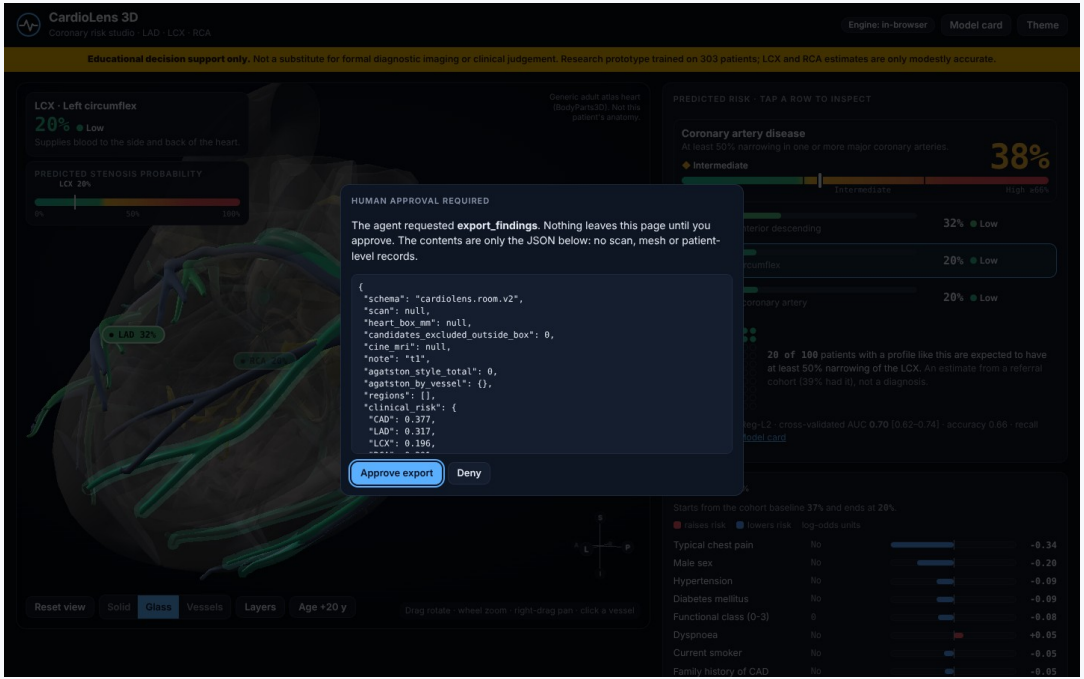




- CT: NIfTI / DICOM / zip, calcium candidates, heart box
- Cine MRI: LV EDV, ESV, EF, CO → beating LV in 3D
- Decoded in the tab, never uploaded
- Automatic estimates, not validated

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WEBMCP

# Agent-native, privacy by design

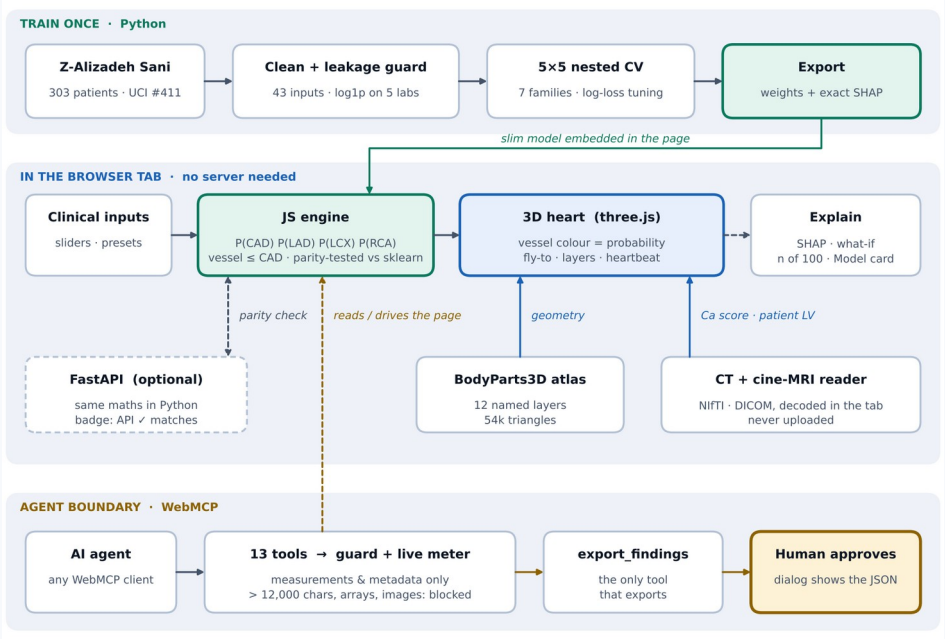



- 13 tools on document.modelContext
- Results are numbers and metadata only; guard blocks arrays and images
- Live meter shows what crossed the boundary
- export_findings waits for human approval

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SYSTEM DESIGN

# Architecture




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TECHNICAL IMPLEMENTATION

# Technical stack and quality

- Python: pandas, scikit-learn, SHAP, FastAPI (optional API)
- Frontend: vanilla JavaScript, three.js r128, BodyParts3D atlas, WebMCP
- Packaging: one self-contained HTML file; Docker; GitHub Pages / Render
- 490 automated checks: parity, API, scan, room, DICOM, cine, Chromium browser tests
- No third-party requests; CSP restricts the page to its own origin

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HONEST EVALUATION

# Limits and next steps

- Single-centre data (303 patients, 71% CAD); no external validation
- LCX (0.70) and RCA (0.73) are only weakly predictable
- Generic adult atlas, not patient-specific anatomy
- CT score and MRI estimates are unvalidated demonstrations
- Next: external validation, whole-heart segmentation, clinician study

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PROJECT RESOURCES

# Links

- GitHub repository: [URL]
- Live demo: [URL]
- Demo video (YouTube, unlisted): https://youtu.be/bZg4DH5eqr8
- Contact: theharshyadhav@gmail.com